

Glow Haven Beauty Studio Website – POE Part 3 Improvements

WEDE5020 – ST10516068

KARABO MOJALEFA

INTRODUCTION

This document outlines the improvements that were made to the Glow Haven Beauty Studio website when upgrading to the project from Part 2 to Part 3. The purpose of these improvements was to make the website more interactive, improve the user experience, add functionality through JavaScript, and improve visibility on search engines using Search Engine Optimization (SEO) techniques.

SEARCH ENGINE OPTIMISATION (SEO)

SEO was added to all web pages to improve the visibility of the website on search engines. Meta tags were included inside the head section of each page.

Example of SEO tags used:

- `<meta name="description" content="Glow Haven Beauty Studio offers hair, nail, skincare and makeup services in Johannesburg.">`
- `<meta name="keywords" content="beauty salon, nails, makeup, skincare, hair salon, Johannesburg">`
- `<meta name="author" content="Karabo Mojalefa">`

The description tag provides information about the web page content. The keywords tag identifies important search terms related to the website. The author tag identifies the creator of the web page.

JAVASCRIPT AND FORM VALIDATION

An external JavaScript file was created and linked to the website pages. JavaScript was used to add functionality and improve user interaction. Using an external JavaScript file improves code organisation, reduces duplication, and makes website maintenance easier.

The Contact form validates:

- Full name field
- Email address format.
- Phone number length.
- Message Type selection.
- Minimum message length

This prevents incomplete or incorrect form submissions and improves data quality.

The Booking form validates:

- Customer name is entered.
- Valid email address is provided.
- Age is greater than zero.
- A service is selected.
- A booking date is selected.

This helps prevent invalid bookings and improves user experience. The booking system was also updated to only allow future dates to be selected.

The Enquiry form validates:

- Customer name
- Selected service
- Preferred booking date

This ensures users provide all required information before submitting enquiries. When forms are submitted successfully, a confirmation message is shown to the user.

SEARCH FUNCTIONALITY

A search feature was added to the Services page. Users can enter keywords into the search bar, and the table automatically filters matching services. This allows visitors to quickly locate information without scrolling through the entire page and improves website usability.

Users can search for specific services such as:

- Hair Cut
- Acrylic Nails
- Pedicure
- Facials
- Bridal Makeup

GALLERY LIGHTBOX

A lightbox feature was added to the Gallery page. Users can click on an image to view a larger version of the picture. This improves user experience by allowing visitors to view images in more detail without leaving the page.

When users click on an image:

- The image open sin a larger overlay
- A close (X) button appears
- Users can close the enlarged image and return to the gallery

This provides a more professional and interactive gallery experience

GOOGLE MAPS INTEGRATION

An embedded Google Map was added to the Contact page using an iframe element. This allows visitors to view the physical location of the business directly from the website. Users can zoom, navigate, and obtain directions using the interactive map.

RESPONSIVE MOBILE DESIGN

Additional CSS media queries were implemented to improve responsiveness on smaller screens.

Improvements include:

- Mobile-friendly navigation menu
- Responsive images
- Smaller gallery images on mobile devices
- Improved content layout for smartphones and tablets

The website provides a better experience across multiple screen sizes.

ROBOTS.TXT AND XML SITEMAP

A robots.txt file was created to guide search engine crawlers when indexing the website. An XML sitemap was created to provide search engines with a structured list of webpages contained within the website.

CONCLUSION

The website was successfully improved from Part 2 to Part 3 through the implementation of SEO techniques, JavaScript validation, search functionality, gallery interactivity, Google Maps integration, and search engine indexing tools. These improvements increased the functionality, usability, and professionalism of the website.

REFERENCES

- [1] Javacodepoint (2024). Searching Algorithms in Java with Examples - Javacodepoint. [online] Javacodepoint. Available at: <https://javacodepoint.com/logical-programs/searching-algorithms-in-java-with-examples/>.
- [2] Artist, C. (2024). Image Gallery with Lightbox Effect Using HTML, CSS, and JavaScript | Coding Artist. [online] Coding Artist. Available at: <https://codingartistweb.com/2024/12/image-gallery-with-lightbox-effect-using-html-css-and-javascript/> [Accessed 17 Jun. 2026].
- [3] W3Schools (2019). JavaScript Form Validation. [online] W3schools.com. Available at: https://www.w3schools.com/js/js_validation.asp.
- [4] Usepattern.com. (2025). How to Implement SEO: A Step-by-Step Guide for Beginners. [online] Available at: <https://www.usepattern.com/resources/how-to-implement-seo#understanding-keywords> [Accessed 17 Jun. 2026].
- [5] W3Schools (2019). How To Build a Website. [online] W3schools.com. Available at: https://www.w3schools.com/howto/howto_website.asp.